

IMAT MOCK EXAM 4 — "Calculation Spine"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 43/90.0 · Mock exam · answer key + worked solutions at the end

Section A - Logical Reasoning & Reading (Q1-9)

1. "Osmosis is the net movement of water across a selectively permeable membrane from a region of lower solute concentration to a region of higher solute concentration. In animal cells, an extracellular solution that is more concentrated than the cytoplasm causes the cell to lose water and shrink; a more dilute solution causes water to enter and the cell to swell."

Which is consistent with the passage?

- A) In a dilute external solution, an animal cell shrinks.
- B) Osmosis requires ATP.
- C) Water moves from high to low solute concentration.
- D) Water moves toward the higher-solute side.
- E) A concentrated external solution causes animal cells to swell.

2. "In genetics, a test cross is used to determine whether an individual displaying a dominant phenotype is homozygous or heterozygous. The individual is crossed with a homozygous recessive partner. If any offspring show the recessive phenotype, the tested individual must be heterozygous."

Which is the strongest inference?

- A) Test crosses require a homozygous dominant partner.
- B) A test cross always produces a 3:1 ratio.
- C) If all offspring show the dominant phenotype, the tested individual is definitely homozygous recessive.
- D) Test crosses can reveal the genotype only of recessive individuals.
- E) Observation of any recessive offspring proves the tested individual is heterozygous.

3. (General knowledge) The Italian chemist who proposed a number equal to the number of particles in one mole ($\sim 6.022 \times 10^{23}$), honoured in the constant's name though he did not measure it, is:

- A) Giulio Natta
- B) Amedeo Avogadro
- C) Enrico Fermi
- D) Stanislao Cannizzaro
- E) Mario Molina

4. Six boxes are stacked. Box R is above S but below T. Box U is immediately below S. Box V is at the very top. The order from top to bottom could be:

- A) T, V, R, S, U, W
- B) V, U, S, R, T, W
- C) V, T, R, S, U, (and W)
- D) V, R, T, S, U, W
- E) V, T, S, R, U, W

5. All squares are rectangles. Some rectangles are rhombi. Therefore:

- A) No valid conclusion about squares being rhombi follows.
- B) No square is a rhombus.
- C) Some rectangles are squares.
- D) All squares are rhombi.
- E) Some squares are rhombi.

6. If 5 machines make 5 widgets in 5 minutes, how long does it take 100 machines to make 100 widgets?

- A) 1 minute B) 5 minutes C) 20 minutes D) 100 minutes E) 500 minutes

7. What is the angle between the hands of a clock at 12:30?
A) 0° **B)** 15° **C)** 30° **D)** 165° **E)** 180°
8. Pointing to a photograph, Maria says, "Her mother is my mother's only daughter." How is the person in the photograph related to Maria?
A) Maria herself
B) Maria's mother
C) Maria's daughter
D) Maria's aunt
E) Maria's sister
9. A number is doubled, then 8 is added; the result is 20. The number is:
A) 4 **B)** 6 **C)** 8 **D)** 10 **E)** 12

Section B - Biology (Q10-32)

10. The mitochondrial matrix is the site of:
A) Lactate fermentation
B) Glycolysis
C) The light reactions of photosynthesis
D) The electron transport chain
E) The Krebs cycle
11. The net reaction of cellular respiration is:
A) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$
B) $C_6H_{12}O_6 \rightarrow 2 \text{ lactate}$
C) $C_6H_{12}O_6 \rightarrow 2 \text{ ethanol} + 2CO_2$
D) $6O_2 + 6CO_2 \rightarrow C_6H_{12}O_6$
E) $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
12. Which gas is produced by muscle cells during anaerobic respiration in yeast but NOT in humans?
A) CO_2 **B)** CH_4 **C)** N_2 **D)** O_2 **E)** H_2
13. The inner mitochondrial membrane is folded into:
A) Villi **B)** Microvilli **C)** Thylakoids **D)** Cisternae **E)** Cristae
14. The Calvin cycle of photosynthesis occurs in the:
A) Cytosol
B) Thylakoid membrane
C) Matrix
D) Stroma
E) Nucleus
15. Which is the correct site of the light-dependent reactions?
A) Stroma
B) Matrix
C) Thylakoid membrane
D) Cell wall
E) Cytosol
16. A solution of pH 2 is how many times more acidic than a solution of pH 5?
A) 3 times **B)** 10 times **C)** 30 times **D)** 100 times **E)** 1000 times

- 17.** The four levels of protein structure in order are:
A) Quaternary, tertiary, secondary, primary
B) Primary, tertiary, secondary, quaternary
C) Primary, secondary, quaternary, tertiary
D) Primary, secondary, tertiary, quaternary
E) Secondary, primary, tertiary, quaternary
- 18.** Which monosaccharide is the primary energy source for cells and the product of dietary carbohydrate digestion?
A) Fructose **B)** Galactose **C)** Glucose **D)** Lactose **E)** Sucrose
- 19.** Which of the following is a disaccharide?
A) Sucrose **B)** Glycogen **C)** Cellulose **D)** Fructose **E)** Glucose
- 20.** Lipids are largely insoluble in water because they are:
A) Highly polar
B) Non-polar (hydrophobic)
C) Composed only of amino acids
D) Charged
E) Always small molecules
- 21.** DNA differs from RNA in that DNA:
A) Has fewer nucleotide types
B) Is found only in the cytoplasm
C) Contains the sugar ribose
D) Contains thymine instead of uracil
E) Is usually single-stranded
- 22.** A population is in Hardy-Weinberg equilibrium for a locus with two alleles A (frequency p) and a (frequency q). The frequency of the heterozygote Aa is:
A) q^2 **B)** $2pq$ **C)** 1 **D)** $p + q$ **E)** p^2
- 23.** ABO blood groups are an example of:
A) Polygenic inheritance
B) Sex-linked inheritance
C) X-inactivation
D) Mitochondrial inheritance
E) Codominance (and multiple alleles)
- 24.** In humans, the normal diploid number of chromosomes is:
A) 23 **B)** 44 **C)** 46 **D)** 48 **E)** 92
- 25.** The cell-cycle phase in which DNA is replicated is:
A) S phase **B)** Cytokinesis **C)** G2 **D)** M phase **E)** G1
- 26.** Which vessels have thick muscular walls and carry blood away from the heart under high pressure?
A) Venules **B)** Arteries **C)** Veins **D)** Lymph vessels **E)** Capillaries
- 27.** The exchange of O_2 and CO_2 between blood and air occurs by:
A) Phagocytosis
B) Exocytosis
C) Active transport
D) Osmosis
E) Diffusion (passive)

- 28.** The hormone secreted by the adrenal medulla in the "fight or flight" response is:
A) Adrenaline (epinephrine)
B) Thyroxine
C) Insulin
D) Oestrogen
E) Cortisol
- 29.** The longest bone in the human body is the:
A) Tibia **B)** Humerus **C)** Femur **D)** Spine **E)** Skull
- 30.** Smooth muscle is characterised by:
A) Multinucleation
B) Presence in the heart only
C) Branched cells with intercalated discs
D) Lack of striation and involuntary control
E) Striation and voluntary control
- 31.** The organelle responsible for ATP production in animal cells is the:
A) Mitochondrion **B)** Nucleus **C)** Chloroplast **D)** Lysosome **E)** Ribosome
- 32.** Which is the correct sequence of blood flow through the heart, starting from the body?
A) Vena cava → left atrium → left ventricle → aorta
B) Vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta
C) Aorta → right atrium → right ventricle → pulmonary vein
D) Right atrium → left atrium → right ventricle → left ventricle
E) Pulmonary vein → right atrium → right ventricle → pulmonary artery

Section C - Chemistry (Q33-47)

- 33.** Calculate the number of moles in 22 g of CO₂. (C = 12, O = 16.)
A) 0.25 **B)** 0.5 **C)** 1 **D)** 2 **E)** 22
- 34.** What volume does 0.25 mol of an ideal gas occupy at STP (V_m = 22.4 L/mol)?
A) 2.8 L **B)** 5.6 L **C)** 11.2 L **D)** 22.4 L **E)** 44.8 L
- 35.** How many atoms are in 1.5 mol of He?
A) 1.20×10^{24} **B)** 1.50×10^{23} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 9.03×10^{23}
- 36.** What is the molarity of a solution made by dissolving 5.85 g NaCl (molar mass 58.5) in water to make 500 mL?
A) 0.1 M **B)** 0.2 M **C)** 0.5 M **D)** 1.0 M **E)** 2.0 M
- 37.** The oxidation number of Cr in Cr₂O₇²⁻ is:
A) -2 **B)** +3 **C)** +6 **D)** +7 **E)** +12
- 38.** Identify the oxidising agent in: CuO + H₂ → Cu + H₂O.
A) Cu
B) H₂O
C) H₂ (it is oxidised)
D) CuO (it is reduced)
E) None
- 39.** Calculate the pH of a 0.05 M solution of a strong base NaOH (assume full dissociation, 25 °C).
A) 1.3 **B)** 7 **C)** 12.7 **D)** 13 **E)** 14
- 40.** The conjugate base of H₂O is:
A) H⁺ **B)** O₂ **C)** H₃O⁺ **D)** OH⁻ **E)** H₂

41. Which has the greatest first ionisation energy?
A) Ar **B)** Cl **C)** Na **D)** Al **E)** Mg
42. Which molecule has a linear geometry?
A) CH₄ **B)** H₂O **C)** SO₂ **D)** NH₃ **E)** CO₂
43. Which intermolecular force is strongest?
A) Hydrogen bonding
B) London dispersion
C) Van der Waals (general)
D) Dipole-dipole
E) Covalent bonding (intramolecular)
44. Increasing pressure on a gas at constant temperature will (Boyle's law):
A) Increase its volume
B) Increase its temperature
C) Decrease its volume
D) Have no effect
E) Decrease its moles
45. The structural formula CH₃-CH₂-OH represents:
A) Ethanoic acid **B)** Diethyl ether **C)** Propanol **D)** Ethanol **E)** Methanol
46. In the reaction: AgNO₃ + NaCl → AgCl + NaNO₃, the precipitate is:
A) NaCl
B) None — no precipitate forms
C) NaNO₃
D) AgNO₃
E) AgCl (insoluble)
47. For a reaction at equilibrium, adding a catalyst will:
A) Stop the reaction
B) Shift equilibrium toward reactants
C) Shift equilibrium toward products
D) Decrease K_{eq}
E) Increase the rate of both forward and reverse equally; equilibrium position unchanged

Section D - Mathematics (Q48-54)

48. Solve: $2x^2 - 8 = 0$.
A) No real solution
B) $x = \pm 4$
C) $x = \pm 2$
D) $x = 4$ only
E) $x = 2$ only
49. Solve: $|x - 3| = 5$.
A) $x = 2$
B) $x = -2$ only
C) $x = 8$ only
D) $x = 8$ or $x = -2$
E) $x = 5$
50. Solve: $\log_2(x) + \log_2(x - 3) = 2$ ($x > 3$).
A) $x = 4$ **B)** $x = -1$ **C)** $x = 3$ **D)** $x = 8$ **E)** $x = 1$

51. Solve the inequality: $x^2 - 4 < 0$.

- A) $x < -2$ B) $x < 2$ C) $x \neq 2$ D) $x > 2$ E) $-2 < x < 2$

52. A right circular cylinder has radius 3 and height 10. What is its volume? (Use $\pi = 3.14$.)

- A) 30 B) 94.2 C) 282.6 D) 314 E) 942

53. $\tan 45^\circ = ?$

- A) 1 B) $\sqrt{3}$ C) 0 D) $\sqrt{3}/3$ E) $1/2$

54. A die is rolled once. What is the probability of an even number greater than 2?

- A) $1/6$ B) $1/3$ C) $1/2$ D) $2/3$ E) $5/6$

Section E - Physics (Q55-60)

55. A car accelerates from 5 m/s to 25 m/s in 10 s. What is its acceleration?

- A) 1 m/s^2 B) 2 m/s^2 C) 5 m/s^2 D) 20 m/s^2 E) 25 m/s^2

56. A 12 V battery drives a current through a 4Ω resistor in series with a 2Ω resistor. What is the current?

- A) 1 A B) 2 A C) 3 A D) 6 A E) 12 A

57. A 3 A current flows for 2 minutes. What total charge passes a point?

- A) 6 C B) 60 C C) 180 C D) 360 C E) 600 C

58. A 2 kg object moving at 3 m/s collides with a stationary 1 kg object and they stick together. What is their common velocity?

- A) 1 m/s B) 2 m/s C) 3 m/s D) 4 m/s E) 6 m/s

59. A 4 kg object is dropped (from rest). Ignoring air resistance, what is its velocity after 3 s? ($g = 9.8 \text{ m/s}^2$.)

- A) 9.8 m/s B) 19.6 m/s C) 29.4 m/s D) 39.2 m/s E) 88.2 m/s

60. A wave has frequency 50 Hz and wavelength 4 m. What is its speed?

- A) 12.5 m/s B) 50 m/s C) 100 m/s D) 200 m/s E) 400 m/s

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	D	2	E	3	B	4	C	5	A
6	B	7	D	8	C	9	B	10	E
11	A	12	A	13	E	14	D	15	C
16	E	17	D	18	C	19	A	20	B
21	D	22	B	23	E	24	C	25	A
26	B	27	E	28	A	29	C	30	D
31	A	32	B	33	B	34	B	35	E
36	B	37	C	38	D	39	C	40	D
41	A	42	E	43	A	44	C	45	D
46	E	47	E	48	C	49	D	50	A
51	E	52	C	53	A	54	B	55	B
56	B	57	D	58	B	59	C	60	D

Worked solutions

Each solution shows the reasoning that produces the key, and names the wrong options by their content. Letters refer to the option order printed in this paper.

Question 1 • Answer: D

The passage states that osmosis moves water from lower to higher solute concentration, so the consistent statement is that water moves toward the higher-solute side. The claim that water moves from high to low solute concentration reverses the stated direction, and the claim that a dilute external solution makes an animal cell shrink reverses the stated effect: dilute outside means water enters and the cell swells. Osmosis is passive, so the ATP claim also contradicts the passage.

Question 2 • Answer: E

A recessive offspring must receive a recessive allele from each parent, so a dominant-phenotype parent that produces one must carry the recessive allele, meaning it is heterozygous; this direct restatement of the passage is the strongest inference. The 3:1 ratio belongs to a self-cross of two heterozygotes, not to a test cross (heterozygote $\times$ recessive gives roughly 1:1). The claim that all-dominant offspring prove the parent is homozygous recessive is self-contradictory, and the test-cross partner is homozygous recessive, not dominant.

Question 3 • Answer: B

The constant is Avogadro's number, $\approx 6.022 \times 10^{23}$ particles per mole, named after the Italian chemist Amedeo Avogadro, whose 1811 hypothesis that equal gas volumes contain equal numbers of molecules underpins the mole concept. Cannizzaro later championed Avogadro's ideas but the constant does not bear his name, and Molina, Fermi and Natta are 20th-century scientists unrelated to the mole.

Question 4 • Answer: C

V must be at the very top, and the chain T above R above S, with U immediately below S, fixes the run T, R, S, U in positions 2 to 5, leaving the sixth box W at the bottom: V, T, R, S, U, W. Every other listing breaks a rule: one starts with T, leaving V below the top; one puts R above T; one swaps R and S; and one places U second, contradicting U immediately below S. The arrangement is unique.

Question 5 • Answer: A

The premises connect squares to rectangles (all) and rectangles to rhombi (some), but the 'some rectangles' that are rhombi need not include any square, and nothing forces every square to be a rhombus either; both cases remain logically possible, so no conclusion about squares and rhombi is entailed. The suggestion that some rectangles are squares would require assuming squares exist (existential import), which a universal statement alone does not provide. Although in real geometry a square is in fact a rhombus, that is outside knowledge, not a deduction from these premises.

Question 6 • Answer: B

Step 1: from 5 machines making 5 widgets in 5 minutes, each machine makes 1 widget in 5 minutes. Step 2: 100 machines working in parallel each make 1 widget in the same 5 minutes, so 100 widgets take 5 minutes. The intuitive answer of 100 minutes wrongly scales time with the machine and widget counts, which here grow together and leave the time unchanged.

Question 7 • Answer: D

Step 1: at 30 minutes the minute hand sits at $30/60 \times 360^\circ = 180^\circ$. Step 2: the hour hand moves 360° in 12 hours, i.e. 30° per hour, so in 30 minutes it moves 15° away from the 12. Step 3: the angle between the hands is $180^\circ - 15^\circ = 165^\circ$. The tempting 180° assumes the hour hand is still exactly on 12 and ignores its drift toward 1.

Question 8 • Answer: C

My mother's only daughter is Maria herself, since she has no sisters. The sentence therefore says the photographed person's mother is Maria, so the person in the photograph is Maria's daughter. Maria herself is ruled out because that person's mother would then be Maria's mother, not Maria, and a sister or aunt would require another daughter of Maria's mother.

Question 9 • Answer: B

Let the number be x . Doubling then adding 8 gives $2x + 8 = 20$, so $2x = 12$ and $x = 6$; check: $2 \times 6 + 8 = 20$. Wrong values such as halving 20 directly come from undoing the operations in the wrong order; the correct reversal is subtract 8 first, then halve.

Question 10 • Answer: E

The matrix is the innermost compartment of the mitochondrion and contains the Krebs (citric acid) cycle enzymes. Glycolysis runs in the cytosol, the electron transport chain is embedded in the inner membrane rather than the matrix, lactate fermentation is cytosolic, and the light reactions belong to chloroplast thylakoids, not mitochondria.

Question 11 • Answer: A

Aerobic respiration consumes one glucose with six oxygen molecules and yields six CO_2 , six H_2O and energy: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy}$. The equation $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ runs the other way and is photosynthesis, while $\text{glucose} \rightarrow 2 \text{ lactate}$ and $\text{glucose} \rightarrow 2 \text{ ethanol} + 2\text{CO}_2$ are only the anaerobic fermentation branches, not the overall aerobic reaction.

Question 12 • Answer: A

Yeast ferments pyruvate to ethanol plus CO_2 , so it releases carbon dioxide; human muscle reduces pyruvate to lactate and releases no CO_2 . CO_2 is therefore produced in yeast anaerobic respiration but not in human muscle. O_2 , N_2 , CH_4 and H_2 are products of neither pathway.

Question 13 • Answer: E

The inner mitochondrial membrane folds into cristae, which enlarge the surface available for electron transport chain complexes and ATP synthase. Thylakoids are chloroplast membranes, cisternae are flattened Golgi/ER sacs, and microvilli and villi are folds of cell surfaces or the intestine, none of them mitochondrial.

Question 14 • Answer: D

The Calvin cycle, i.e. the light-independent fixation of CO_2 , takes place in the stroma, the fluid surrounding the thylakoids. The thylakoid membrane hosts the light-dependent reactions instead, and the matrix is a mitochondrial compartment, so neither fits.

Question 15 • Answer: C

The light-dependent reactions occur on the thylakoid membranes, where the photosystems, the electron transport chain and ATP synthase are located. The stroma instead hosts the Calvin cycle, and the cytosol, matrix and cell wall are not sites of the light reactions.

Question 16 • Answer: E

Each pH unit corresponds to a factor of 10 in H^+ concentration, so a difference of $5 - 2 = 3$ units gives a ratio of $10^3 = 1000$. The answer 30 comes from multiplying the difference by 10 as if the scale were linear, and 100 from taking the difference as 2 units.

Question 17 • Answer: D

The levels build up as primary (amino acid sequence), secondary (local α -helix and β -sheet held by backbone hydrogen bonds), tertiary (overall 3D fold of one polypeptide) and quaternary (assembly of several chains). Listings that swap secondary and tertiary or reverse the whole order misplace these levels.

Question 18 • Answer: C

Glucose is the monosaccharide used directly in respiration and the main product of dietary carbohydrate (starch and glycogen) digestion. Fructose and galactose are other monosaccharides that must first be converted into glycolytic intermediates, while sucrose and lactose are disaccharides and not monosaccharides at all.

Question 19 • Answer: A

Sucrose consists of glucose joined to fructose by a glycosidic bond, making it a disaccharide. Glucose and fructose themselves are monosaccharides, and cellulose and glycogen are long-chain polysaccharides.

Question 20 • Answer: B

Lipids are built around long non-polar hydrocarbon chains, so water cannot form favourable interactions with them: they are hydrophobic and hence water-insoluble. Charged or highly polar species dissolve readily, so those claims go the wrong way, and lipids are neither amino-acid based nor necessarily small molecules.

Question 21 • Answer: D

The decisive fact is that DNA uses the base thymine where RNA uses uracil. The same contrast rules out the other claims: DNA contains deoxyribose rather than ribose, DNA is usually double-stranded (RNA is the usually single-stranded one), both use four types of nucleotide, and DNA resides chiefly in the nucleus, not the cytoplasm.

Question 22 • Answer: B

In Hardy-Weinberg equilibrium the genotype frequencies are p^2 for AA, $2pq$ for the heterozygote Aa and q^2 for aa, with $p + q = 1$. The heterozygote frequency is therefore $2pq$; p^2 and q^2 are the two homozygotes, and $p + q$ is simply 1.

Question 23 • Answer: E

Type AB individuals express both the A and B antigens, showing codominance, and the system involves three alleles (I^A , I^B , i) in the population, i.e. multiple alleles. The trait is autosomal, so sex-linked inheritance, mitochondrial inheritance, polygenic inheritance and X-inactivation do not describe it.

Question 24 • Answer: C

Human somatic (diploid) cells carry $2n = 46$ chromosomes, arranged as 23 pairs. The value 23 is the haploid (gamete) number, and 44 counts only the 22 autosome pairs while omitting the two sex chromosomes.

Question 25 • Answer: A

DNA replication occurs in S (synthesis) phase of interphase, when the genome is copied. G1 and G2 are growth phases without replication, M phase is mitosis, and cytokinesis divides the cytoplasm.

Question 26 • Answer: B

Arteries carry blood away from the heart at high pressure and therefore have thick, elastic, muscular walls. Veins and venules return blood at low pressure through thinner walls, capillaries are one-cell-thick exchange vessels, and lymph vessels are also thin-walled and low pressure.

Question 27 • Answer: E

Gas exchange across the alveolar and capillary walls occurs by passive diffusion down partial-pressure gradients: O_2 moves in and CO_2 out with no energy cost. Osmosis concerns water movement, and active transport, phagocytosis and exocytosis are energy-requiring processes not used for respiratory gas exchange.

Question 28 • Answer: A

The adrenal medulla secretes adrenaline (epinephrine), which drives the fight-or-flight response: raised heart rate, blood pressure and blood glucose. Cortisol is made in the adrenal cortex, insulin in the pancreas, thyroxine in the thyroid and oestrogen in the ovaries, so none of them is the medulla's emergency hormone.

Question 29 • Answer: C

The femur, the thigh bone, is the longest and strongest bone in the human body. The tibia and humerus are also long bones but shorter, and the spine and skull are structures made of many bones, not single long bones.

Question 30 • Answer: D

Smooth muscle consists of spindle-shaped cells with no striations and is controlled involuntarily by the autonomic nervous system. Striations plus voluntary control describe skeletal muscle, and branched cells with intercalated discs describe cardiac muscle, as does presence in the heart only.

Question 31 • Answer: A

Mitochondria perform oxidative phosphorylation and supply most of the ATP in animal cells. Chloroplasts do this job only in plants (animal cells have none), ribosomes build proteins, lysosomes digest material and the nucleus stores DNA, so none of them is the ATP-producing organelle.

Question 32 • Answer: B

From the body, blood enters the right side via the vena cava, flows right atrium → right ventricle → pulmonary artery → lungs, returns via the pulmonary vein → left atrium → left ventricle and leaves through the aorta. Shortcuts that start at the aorta or the pulmonary vein, or that route pulmonary-vein blood into the right atrium, contradict this one-way circuit.

Question 33 • Answer: B

Step 1: molar mass of $\text{CO}_2 = 12 + 2 \times 16 = 44 \text{ g/mol}$. Step 2: $n = m \div M = 22 \text{ g} \div 44 \text{ g/mol} = 0.5 \text{ mol}$. Counting only one oxygen (molar mass 16) or inverting the division produces the other numerical options, and 22 is the given mass, not an amount in moles.

Question 34 • Answer: B

At STP one mole of ideal gas occupies 22.4 L, so $V = n \times V_m = 0.25 \times 22.4 = 5.6 \text{ L}$. The value 11.2 L is the volume of 0.5 mol, twice the given amount, and 22.4 L would require a full mole.

Question 35 • Answer: E

Helium is monatomic, so the number of atoms equals moles $\times$ Avogadro's number: $1.5 \times 6.022 \times 10^{23} = 9.03 \times 10^{23}$. The plain 6.02×10^{23} forgets the 1.5 multiplier (that is one mole), and 3.01×10^{23} corresponds to only 0.5 mol.

Question 36 • Answer: B

Step 1: $n = 5.85 \text{ g} \div 58.5 \text{ g/mol} = 0.1 \text{ mol}$. Step 2: convert the volume: $500 \text{ mL} = 0.5 \text{ L}$. Step 3: $M = 0.1 \text{ mol} \div 0.5 \text{ L} = 0.2 \text{ M}$. Stopping after step 1 and quoting the mole amount as a concentration gives the 0.1 M trap.

Question 37 • Answer: C

Let the oxidation number of Cr be x . With oxygen at -2 and the ion's total charge at -2 : $2x + 7 \times (-2) = -2$, so $2x = 12$ and $x = +6$. The value $+12$ stops before dividing by the two chromium atoms, and -2 is the contribution of the oxygens, not of chromium.

Question 38 • Answer: D

Copper goes from $+2$ in CuO to 0 in Cu , so CuO gains electrons and is reduced; the species that is reduced is by definition the oxidising agent. H_2 loses electrons (it is oxidised to H_2O) and is the reducing agent, and the copper metal is the reduction product, not an agent.

Question 39 • Answer: C

Step 1: NaOH dissociates fully, so $[\text{OH}^-] = 0.05 \text{ M}$. Step 2: $\text{pOH} = -\log(0.05) = -\log(5 \times 10^{-2}) = 2 - \log 5 \approx 2 - 0.70 = 1.30$. Step 3: $\text{pH} = 14 - 1.30 = 12.70$. The value 1.3 is exactly this pOH misread as a pH, and 13 would correspond to 0.1 M .

Question 40 • Answer: D

A conjugate base is what remains after the acid donates one proton: $\text{H}_2\text{O} - \text{H}^+ = \text{OH}^-$, so the hydroxide ion is water's conjugate base. H_3O^+ is instead water's conjugate acid, formed when water accepts a proton, and H^+ is the transferred particle itself, not the remaining species.

Question 41 • Answer: A

First ionisation energy rises across Period 3 as nuclear charge pulls on electrons in the same shell, peaking at the noble gas; argon's complete third shell makes it the highest of the five. Sodium is the lowest with its single, easily lost $3s$ electron, and chlorine, though high, still lies below argon.

Question 42 • Answer: E

CO_2 has two electron domains around the central carbon (two double bonds, no lone pairs), which spread to 180° , giving linear $\text{O}=\text{C}=\text{O}$. H_2O is bent (two lone pairs on O), NH_3 is trigonal pyramidal, SO_2 is bent, and CH_4 is tetrahedral with four bonding pairs, so none of them is linear.

Question 43 • Answer: A

Hydrogen bonding, the attraction between H bonded to N, O or F and a lone pair on another electronegative atom, is the strongest of the intermolecular forces listed; dipole-dipole and London dispersion (the van der Waals family) are weaker. Covalent bonding is an intramolecular bond holding atoms within a molecule, so it does not count as an intermolecular force.

Question 44 • Answer: C

Boyle's law, $pV = \text{constant}$ at fixed temperature, makes pressure and volume inversely proportional ($P \propto 1/V$), so increasing pressure decreases the volume. Temperature is constant by assumption and the amount of gas is fixed, so neither the temperature nor the mole number can change.

Question 45 • Answer: D

The chain $\text{CH}_3\text{-CH}_2\text{-OH}$ has two carbons bearing a hydroxyl group, which is ethanol ($\text{C}_2\text{H}_5\text{OH}$). Methanol would have one carbon, propanol three, ethanoic acid would carry a -COOH group instead of -OH , and an ether would have C-O-C without the O-H bond.

Question 46 • Answer: E

Mixing the solutions exchanges ions, and silver chloride is insoluble, so AgCl comes out of solution as a white precipitate while NaNO_3 stays dissolved (sodium salts and nitrates are always soluble). The two reactants are the dissolved starting salts, and a precipitate does indeed form.

Question 47 • Answer: E

A catalyst lowers the activation energy of the forward and reverse reactions by the same amount, so both rates rise equally and equilibrium is simply reached sooner, with the position and K_{eq} unchanged. Any claim of a shift toward products or reactants, or of a changed K_{eq} , contradicts this symmetry, and the reaction does not stop.

Question 48 • Answer: C

Step 1: $2x^2 = 8$. Step 2: $x^2 = 4$. Step 3: $x = +2$ or $x = -2$, since squaring either gives 4. The answer $x = 2$ only drops the negative root, and $x = \pm 4$ comes from skipping the division by 2 before taking the square root.

Question 49 • Answer: D

The equation $|x - 3| = 5$ says x is 5 units from 3, so $x = 3 + 5 = 8$ or $x = 3 - 5 = -2$. Both roots are needed: answering 8 only or -2 only discards one branch of the absolute value.

Question 50 • Answer: A

Step 1: combine the logs: $\log_2[x(x - 3)] = 2$. Step 2: undo the log: $x(x - 3) = 2^2 = 4$. Step 3: $x^2 - 3x - 4 = 0$, so $(x - 4)(x + 1) = 0$ and $x = 4$ or $x = -1$. The domain $x > 3$ rejects $x = -1$ as extraneous, leaving $x = 4$.

Question 51 • Answer: E

$x^2 - 4 < 0$ factors as $(x - 2)(x + 2) < 0$, which holds between the roots, so $-2 < x < 2$. The reversed region $x < -2$ or $x > 2$ solves the opposite inequality $x^2 > 4$, and the one-sided bounds each miss part of the interval.

Question 52 • Answer: C

$V = \pi r^2 h = 3.14 \times 3^2 \times 10 = 3.14 \times 9 \times 10 = 282.6$. The value 94.2 comes from forgetting to square the radius ($3.14 \times 3 \times 10$), and 30 from omitting π altogether.

Question 53 • Answer: A

In a 45-45-90 right triangle the two legs are equal, so $\tan 45^\circ = \text{opposite} \div \text{adjacent} = 1$. The values $\sqrt{3}$ and $\sqrt{3}/3$ are $\tan 60^\circ$ and $\tan 30^\circ$ respectively, and 0 is $\tan 0^\circ$.

Question 54 • Answer: B

The even faces greater than 2 are 4 and 6, giving 2 favourable outcomes out of 6 equally likely faces: $P = 2/6 = 1/3$. Including the face 2, which is not greater than 2, wrongly gives $3/6 = 1/2$.

Question 55 · Answer: B

Acceleration is the change in velocity over time: $a = (25 - 5) \div 10 = 20 \div 10 = 2 \text{ m/s}^2$. The value 20 m/s^2 is the velocity change never divided by the time, and using the final speed alone ($25 \div 10 = 2.5$) is not the acceleration either.

Question 56 · Answer: B

Step 1: in series, resistances add: $R = 4 + 2 = 6 \Omega$. Step 2: $I = V \div R = 12 \div 6 = 2 \text{ A}$. The value 3 A comes from dividing by only the 4Ω resistor and ignoring the 2Ω in series with it.

Question 57 · Answer: D

Charge is current $\times$ time with time in seconds: $t = 2 \text{ min} = 120 \text{ s}$, so $Q = 3 \times 120 = 360 \text{ C}$. Taking the 2 of '2 minutes' as seconds gives the 6 C trap, and forgetting the conversion entirely produces the smaller values.

Question 58 · Answer: B

Total momentum is conserved: before, $p = 2 \text{ kg} \times 3 \text{ m/s} + 1 \text{ kg} \times 0 = 6 \text{ kg}\cdot\text{m/s}$; after, the combined mass is 3 kg moving at v , so $3v = 6$ and $v = 2 \text{ m/s}$. The value 6 m/s mistakes the initial momentum for the final velocity, and 3 m/s ignores the added mass.

Question 59 · Answer: C

For free fall from rest, $v = g \times t = 9.8 \times 3 = 29.4 \text{ m/s}$; the 4 kg mass does not affect the velocity. The values 9.8 m/s and 88.2 m/s correspond to $t = 1 \text{ s}$ and $t = 9 \text{ s}$, and mass would matter only for weight or energy, not for speed.

Question 60 · Answer: D

Wave speed is frequency $\times$ wavelength: $v = f\lambda = 50 \text{ Hz} \times 4 \text{ m} = 200 \text{ m/s}$. The value 12.5 m/s comes from dividing instead of multiplying, inverting the relation.